Online Resource 1

Design, implementation and technical evaluation of AI-CAI Tutor: A novel teacher-governed, curriculum-aligned generative AI tutoring platform for teaching and learning across educational levels in Ghana

Education and Information Technologies

Authors and affiliations: withheld for anonymised peer review

Contents: Figs. S1–S2 (Agile process); Figs. S3–S8 (entity–relationship diagrams); Figs. S9–S11 (additional interfaces); Tables S1–S3 (Agile traceability, component evidence, browser and accessibility checks).

S1 Agile development process

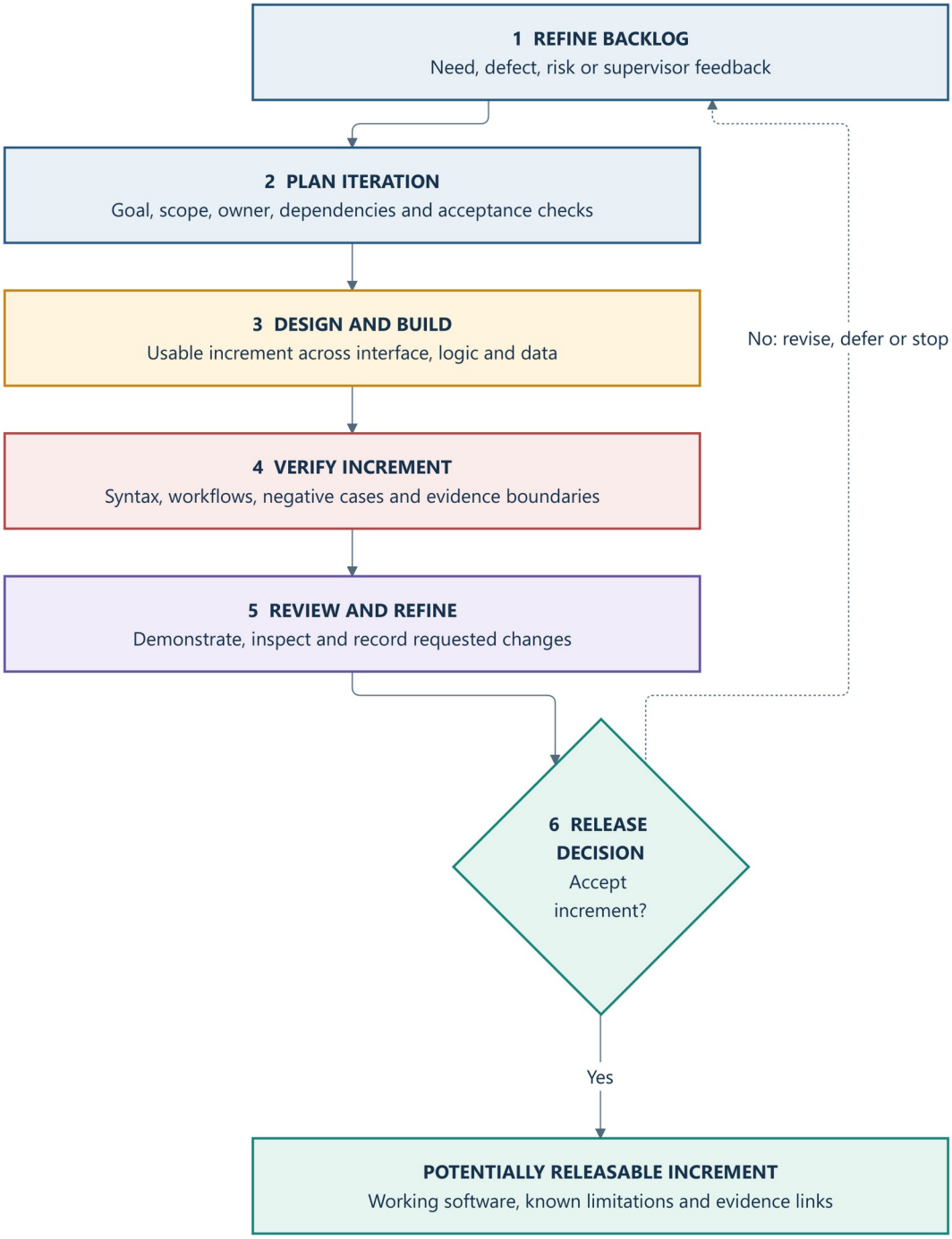


Fig. S1 Agile iteration cycle used to develop the software artefact

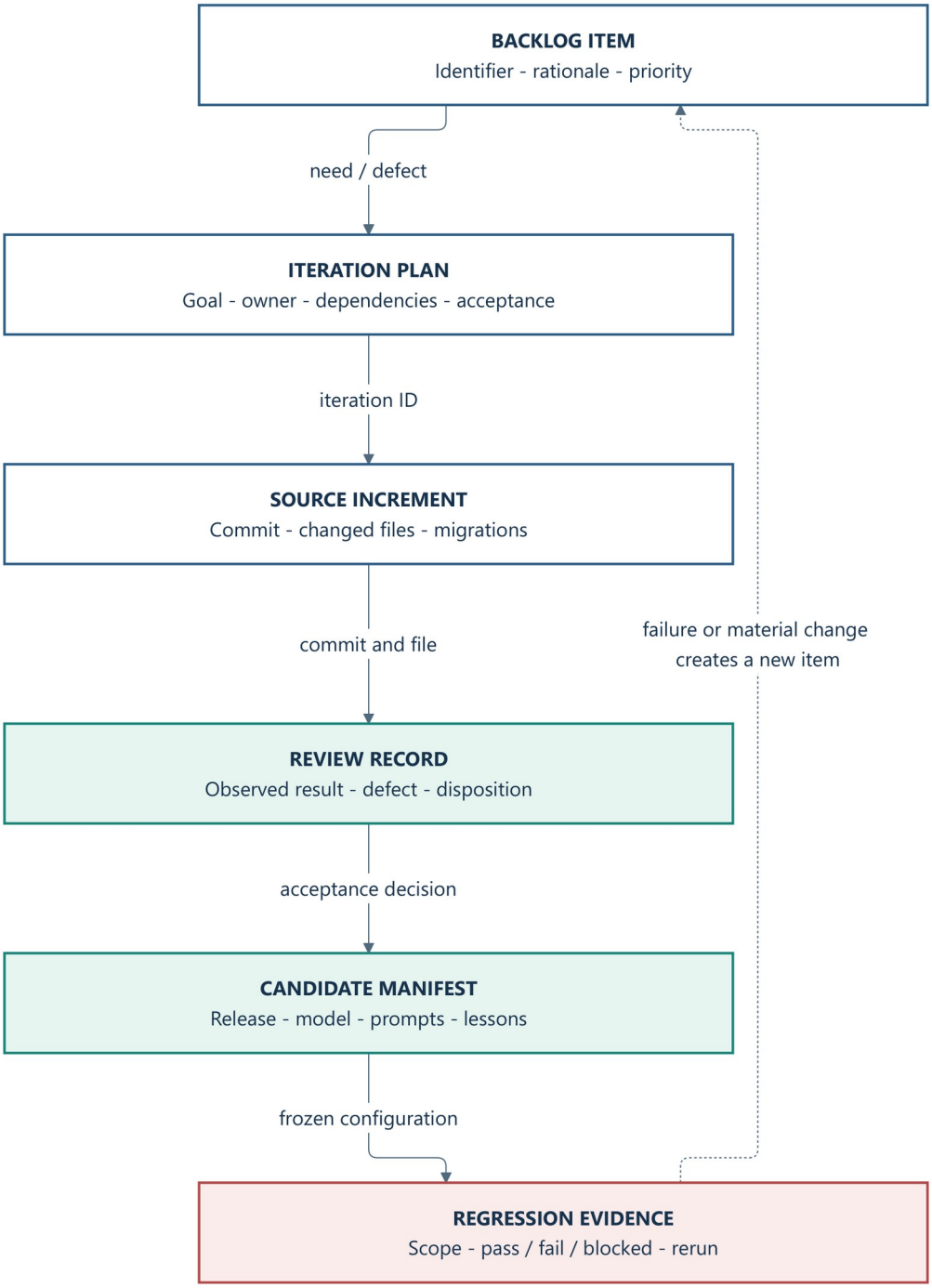


Fig. S2 Agile traceability and release-control chain from backlog item to regression evidence

Table S1 Agile development cycle used for the software artefact and its required traceability

Activity	Project application	Required traceability
Backlog formation	Convert educational needs, implementation defects, safety requirements and feedback into prioritised work items	Identifier, rationale, affected role, risk and acceptance condition
Iteration planning	Select a feasible set of related items and identify technical and content dependencies	Iteration goal, selected items, dependencies, responsible developer and review point

Activity	Project application	Required traceability
Incremental design and build	Implement a usable vertical increment across interface, server logic and persistence where required	Source changes, migrations, configuration changes and affected workflow
Review and refinement	Examine the increment against its acceptance condition and correct identified defects	Reviewer role, observation, requested change, disposition and retest requirement
Release decision	Accept the increment into a candidate, revise it later or stop it when risk prevents acceptance	Commit or release identifier, deferred items, limitations and rerun requirements

S2 Entity–relationship diagrams

All diagrams use crow's-foot notation. PK primary key; FK foreign key.

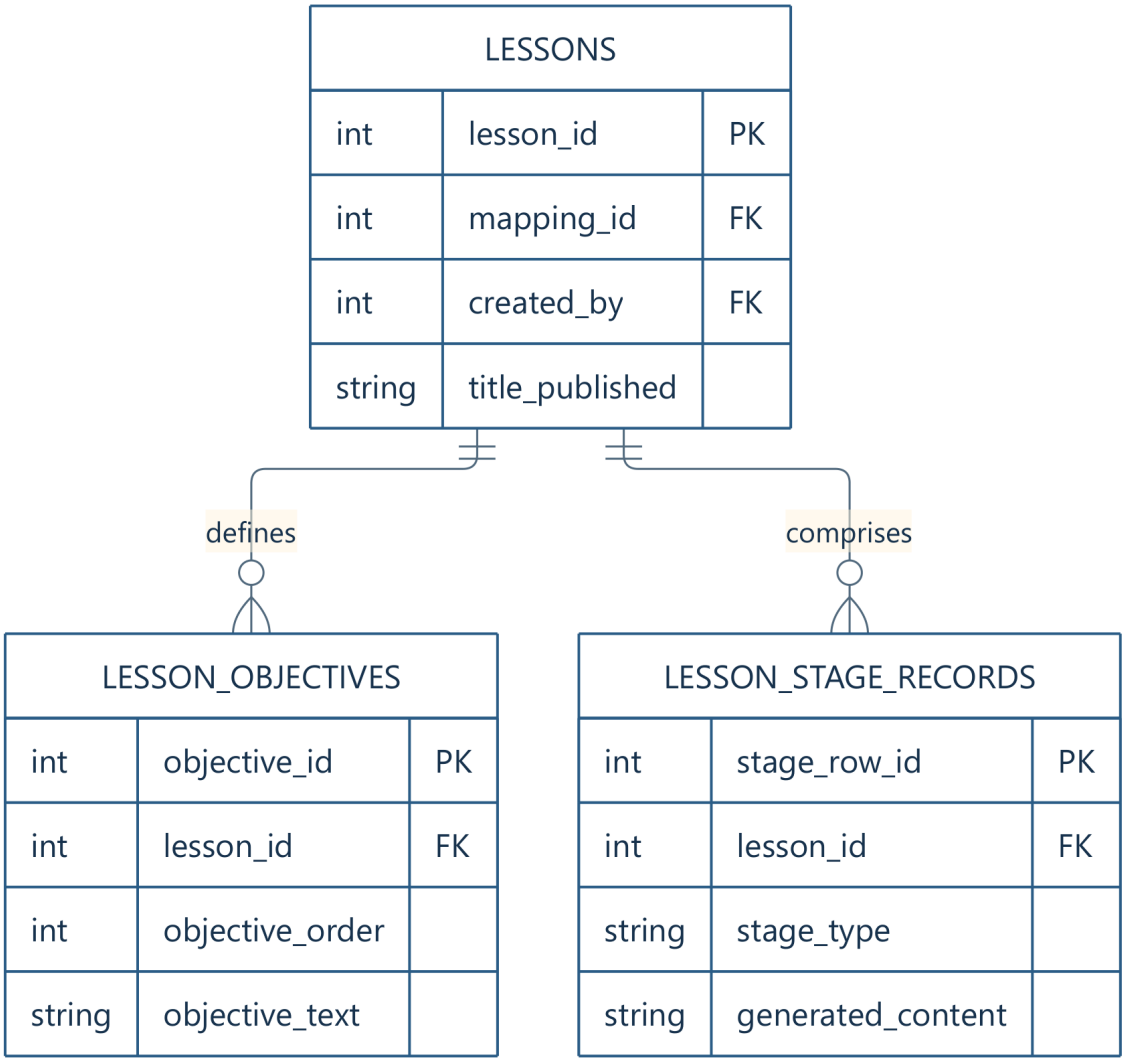


Fig. S3 Entity–relationship diagram for lesson objectives and structured stage records

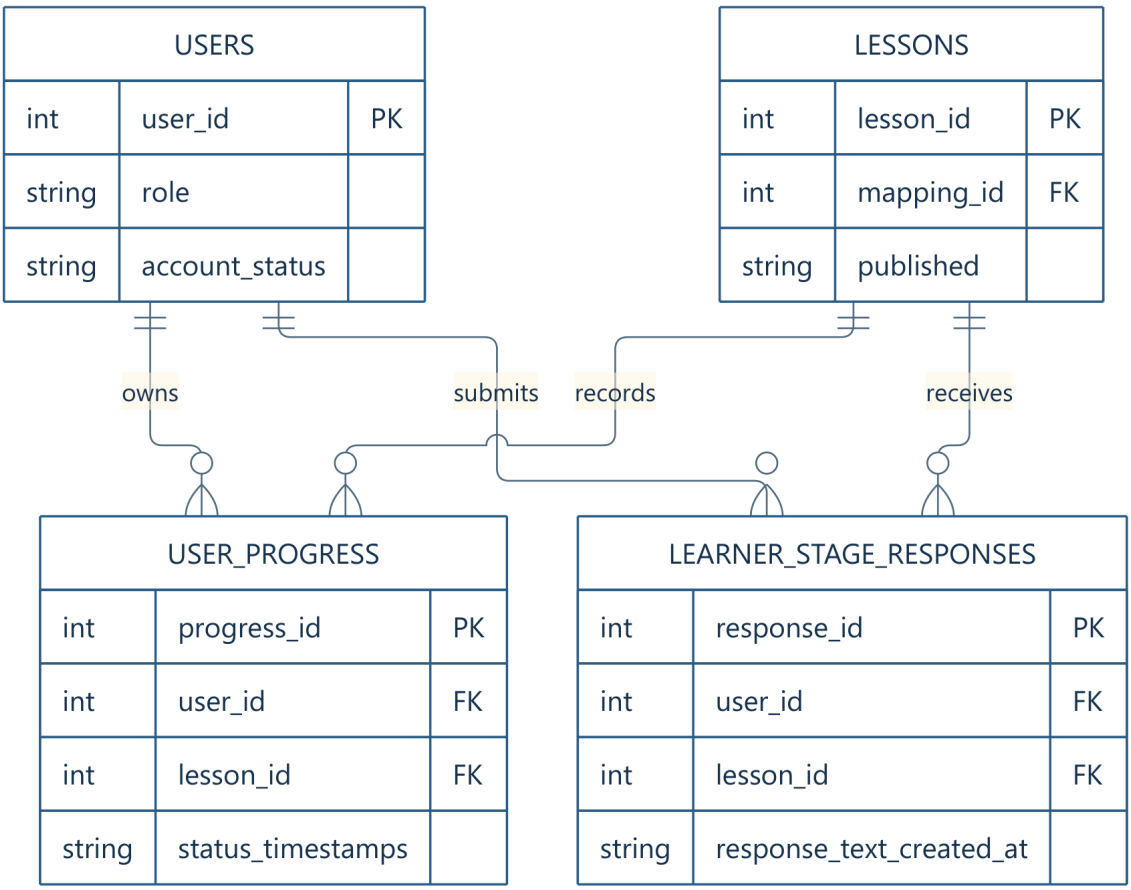


Fig. S4 Entity–relationship diagram for learner progress and lesson-stage responses

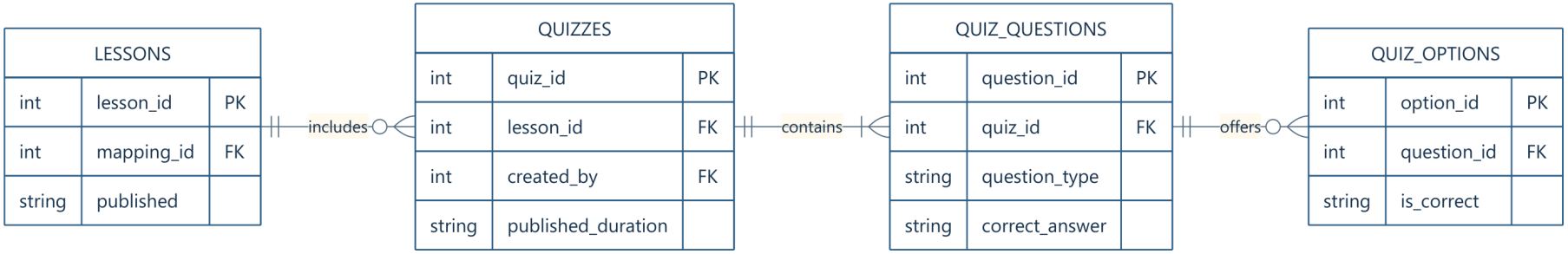


Fig. S5 Entity–relationship diagram for quiz authoring, questions and options

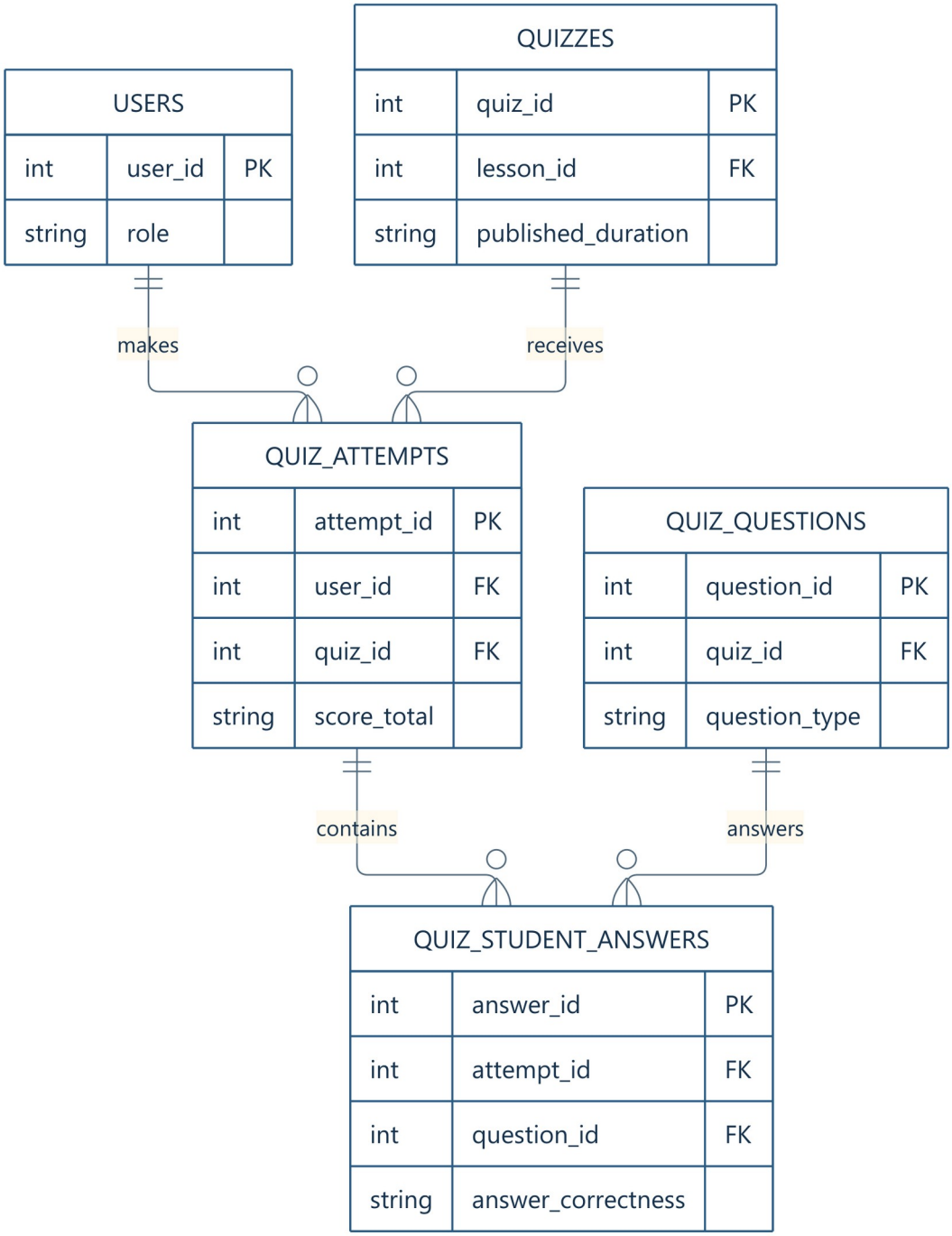


Fig. S6 Entity–relationship diagram for learner quiz attempts and question answers

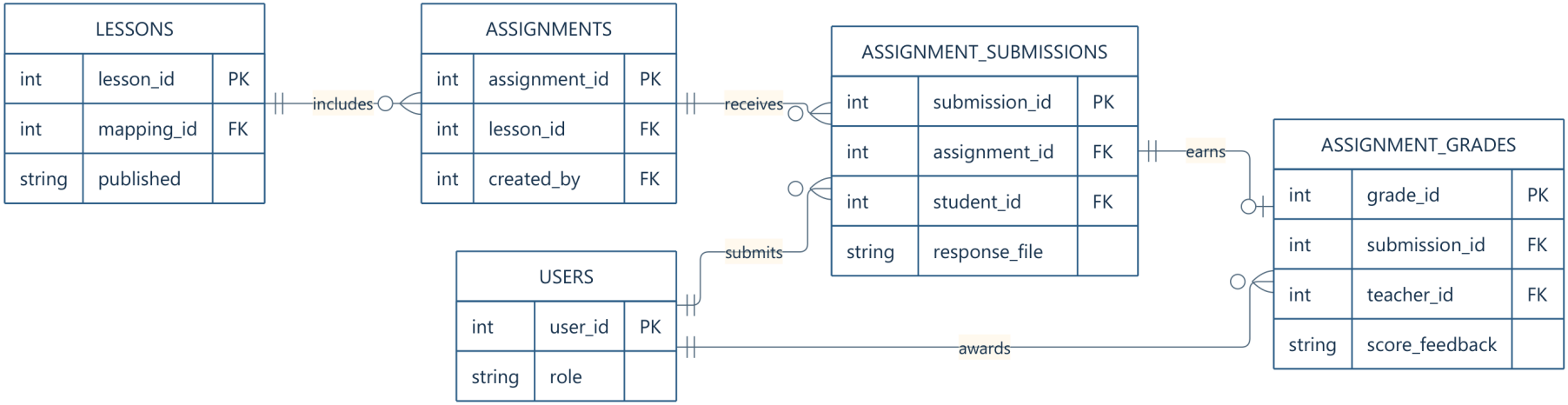


Fig. S7 Entity–relationship diagram for assignments, learner submissions and teacher grades

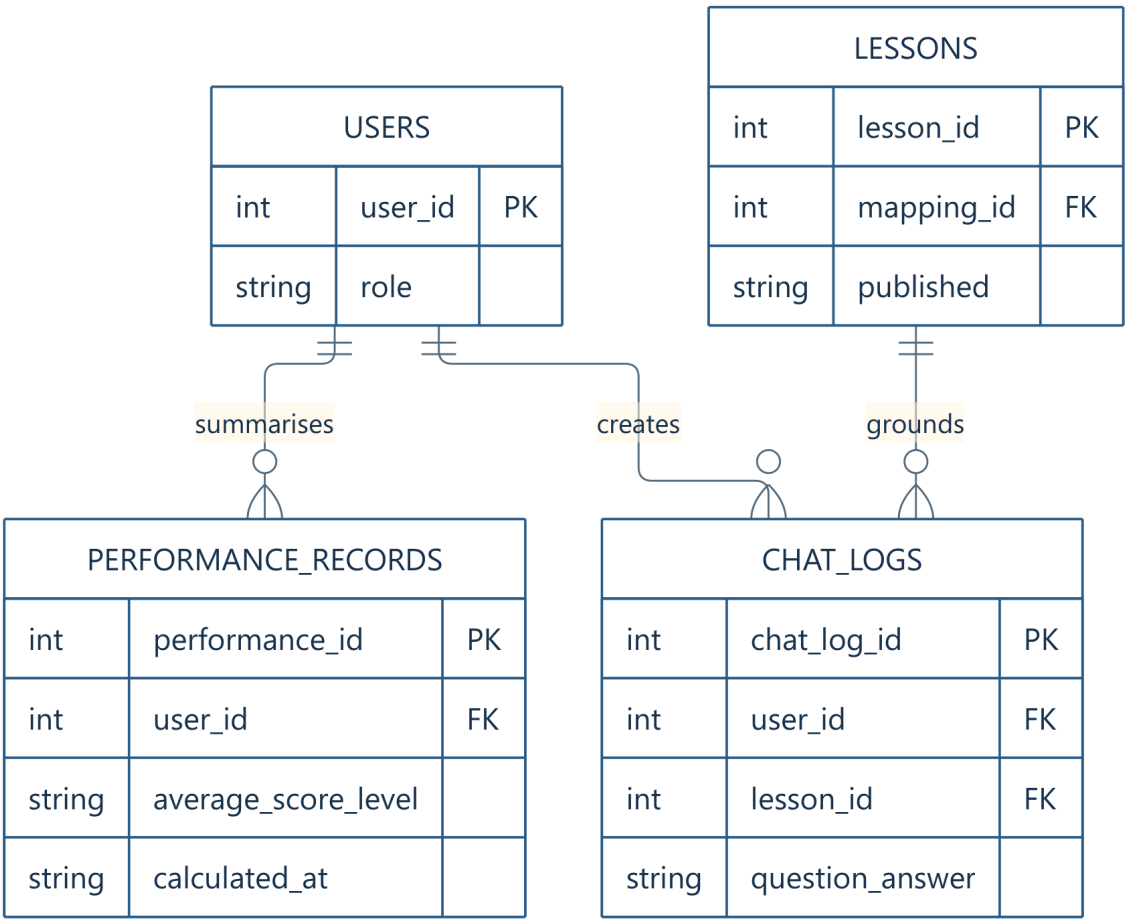


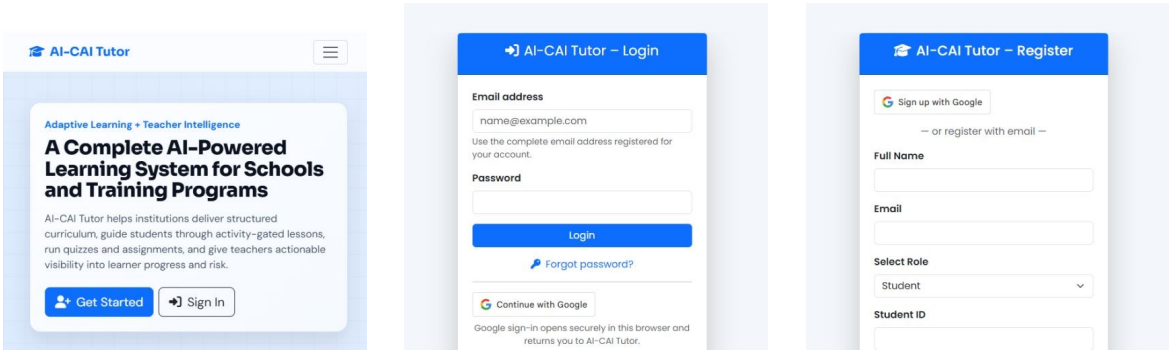
Fig. S8 Entity–relationship diagram for lesson-grounded tutor logs and performance records

S3 Component evidence and additional interfaces

Table S2 Architectural components, responsibilities and representative implementation evidence

Component	Principal responsibility	Representative implementation evidence
Presentation and delivery	Render public, student, teacher and administrator views; provide responsive navigation and installable delivery paths	index.php; role pages under pages/; assets/css/mobile.css; includes/pwa.php; manifest.webmanifest; mobile/
Curriculum service	Create or upload curricula, record level metadata and construct hierarchical mappings	pages/curriculum/create_curriculum.php; upload_curriculum.php; map_curriculum.php
Lesson service	Create structured lesson stages, attachments, publication state and learner-facing lesson access	Curriculum lesson routes; lesson-stage migrations; pages/student/view_lesson.php
Tutor service	Validate access, retrieve active lesson excerpts, construct bounded prompts, moderate content and retain exchanges	includes/tutor_access.php; pages/tutor/chat_handler.php; includes/ai_config.php
Assessment service	Create quizzes and assignments, retain attempts and submissions, support grading and feedback	Routes under pages/assessments/, pages/assignments/ and pages/curriculum/

Component	Principal responsibility	Representative implementation evidence
Monitoring service	Derive progress, performance summaries and teacher-facing activity or risk cues	includes/performance_tracker.php; dashboard and teacher-monitoring routes
Shared controls	Maintain sessions, role authorisation, account approval, route helpers, notifications and provider configuration	includes/auth_guard.php; includes/account_approval.php; includes/google_auth.php; configuration includes
Persistence and adapters	Store operational records and managed files; isolate SQL and external service calls from browser credentials	MySQLi queries, SQL migrations, upload directories and server-side AI, identity and mail handlers



a **b** **c**

Fig. S9 Public presentation and account-entry interfaces: **a** public landing page; **b** login; **c** registration and role selection

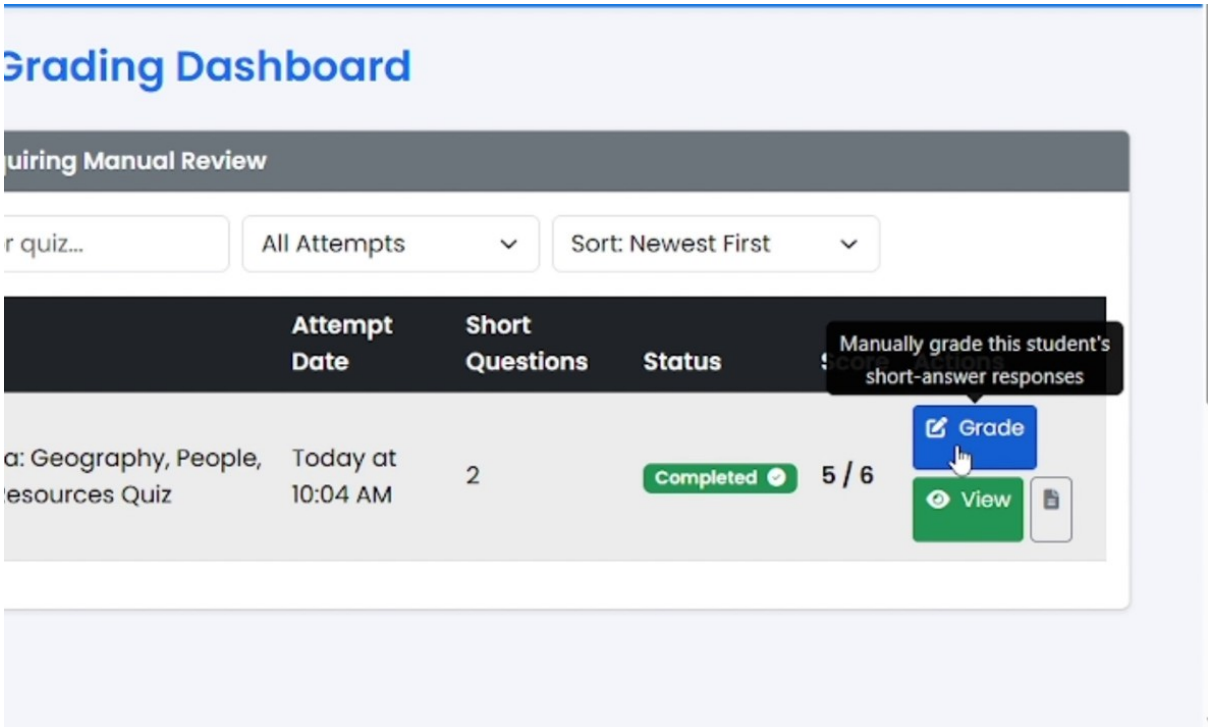


Fig. S10 Teacher-facing manual quiz-grading workflow (identifying columns excluded)

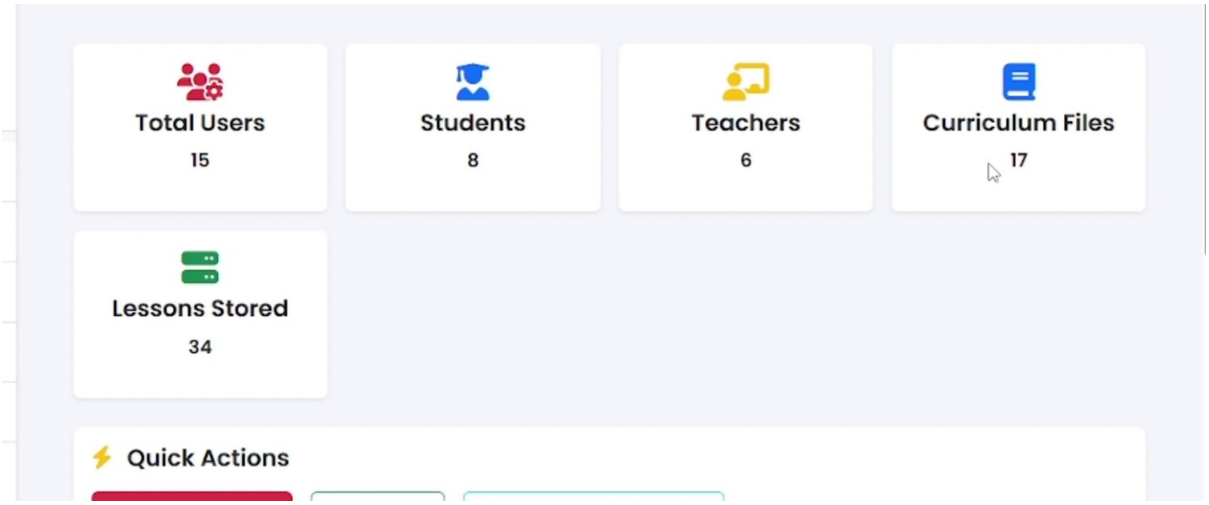


Fig. S11 Administrator operational summary of users, roles, curricula and lessons

S4 Browser and accessibility checks

Table S3 Details of the browser and accessibility checks

Run	Check	Observation
First run (landing, login, registration)	Keyboard traversal of the landing page	27 of 27 expected targets reached
	Visible focus indicator on sampled focus targets	Present on all sampled targets
	Unnamed interactive controls; duplicate identifiers	None on the three routes
	Invalid-login feedback	Exposed as an alert associated with its input
	Landing-page videos	15 of 15 with accessible names and linked text overviews
	Contrast and 320-pixel reflow (sampled pages)	No measured contrast-threshold breach or horizontal overflow
Expanded run (28 October 2026)	axe-core, WCAG 2.0/2.1/2.2 A/AA and best-practice rules	0 violations across 49 visits (10 public routes; 39 authenticated role-route visits)
	Horizontal overflow at 320 CSS pixels, including text-spacing stress	None observed
	Visible button target size	None below 24 × 24 CSS pixels
	Unresolved colour-contrast items	149 items flagged by axe-core for manual review (139 across public routes, mainly the privacy-policy and terms-of-service pages; 10 across seven authenticated visits); not counted as violations
	Media alternatives	Inventory of 16 tutorial and 7 lesson videos, each with a WebVTT caption file and text transcript; role-filtered visits loaded the expected tutorial videos; 1 representative lesson video loaded in the browser; caption accuracy not reviewed by a person